

day 30 | 251110

herein

Adding a startup script to `~/ .ipython/profile_default/startup` is a very handy way to have common imports handled automatically for lab notebook style management.

File naming is weird. Mine is `00-myImports.ipy`. For some reason this is necessary I think. See the README in the directory for more info.

Currently my script is the following:

```
%matplotlib ipynb
```

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
```

This goes just fine for Mac users, but Windows users need to be careful to turn on showing file extensions. You can do this in File Explorer by clicking on View -> More -> Show file extensions.

also our custom library of scripts

For linux users, add the following to `.bashrc` in your home folder. For Mac users, add the same thing to the file `.zshrc`. Note that after `$HOME` your path might be different than mine.

```
# adding ph390 to my PATH but make sure the path is correct
export PYTHONPATH="${PYTHONPATH}:${HOME}/Dropbox/390-fall-2025/scripts/"
```

One way that you can check is by doing this in a jupyter lab cell:

```
import sys
print(sys.path)
```

For Windows users, you need to click Start and type in Environment Variables and select Edit the System Environment Variables. Under user variables add one named PYTHONPATH and give it the value `%USERPROFILE%\Dropbox\390-2025-fall\scripts`. Saving and exiting should do it. Verify by starting jupyter and entering the code above.

to check on things and see whether they are working:

1. just try `np.linspace(1,20)` and see whether that executes without a hiccup.
2. next try `import ph390` and see if that complains. note that this will only work once you have `__inti__.py` and `ph390.py` in that `scripts` folder, even if they are empty. Put a bisection method script in there and see whether you can call it.

hence

We should have done this much earlier in the semester. We don't have to do them all at once, but I think putting the integral methods and the root finding methods and maybe the factorial method in here would have been very helpful at different times. Also doing it would force students to set up their systems like mine and learn how to do this. So this is important and could probably be put in earlier in the semester.

```
In [1]: np.linspace(0,30)
```

```
Out[1]: array([ 0.          ,  0.6122449 ,  1.2244898 ,  1.83673469,  2.44897959,
                3.06122449,  3.67346939,  4.28571429,  4.89795918,  5.51020408,
                6.12244898,  6.73469388,  7.34693878,  7.95918367,  8.57142857,
                9.18367347,  9.79591837, 10.40816327, 11.02040816, 11.63265306,
               12.24489796, 12.85714286, 13.46938776, 14.08163265, 14.69387755,
               15.30612245, 15.91836735, 16.53061224, 17.14285714, 17.75510204,
               18.36734694, 18.97959184, 19.59183673, 20.20408163, 20.81632653,
               21.42857143, 22.04081633, 22.65306122, 23.26530612, 23.87755102,
               24.48979592, 25.10204082, 25.71428571, 26.32653061, 26.93877551,
               27.55102041, 28.16326531, 28.7755102 , 29.3877551 , 30.          ])
```

```
In [2]: from ph390 import *
```

```
In [3]: import sys
        print(sys.path)
```

```
['/home/ehreming/Dropbox/390-2025-fall/scripts', '/home/ehreming/anaconda3/lib/python3.12.zip', '/home/ehreming/anaconda3/lib/python3.12', '/home/ehreming/anaconda3/lib/python3.12/lib-dynload', '', '/home/ehreming/anaconda3/lib/python3.12/site-packages']
```

```
In [4]: import os, sys; os.environ.get("PYTHONPATH"), sys.path[:5]
```

```
Out[4]: ('/home/ehreming/Dropbox/390-2025-fall/scripts',
        ['/home/ehreming/Dropbox/390-2025-fall/scripts',
         '/home/ehreming/anaconda3/lib/python3.12.zip',
         '/home/ehreming/anaconda3/lib/python3.12',
         '/home/ehreming/anaconda3/lib/python3.12/lib-dynload',
         ''])
```

```
In [ ]:
```